STUMPRIDER · RISK REGISTER

StumpRider risk register, hazard analysis and design FMEA

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (SMR-DEC-001 O6 to O9) written in; O10 back with Amish
0.1	2026-10-24	Amish Chadha	First issue in the kit 1.9.1 upgrade; project risks, preliminary hazard analysis and first-pass design FMEA of option A

StumpRider risk register, hazard analysis and design FMEA

The largest hazard is the one the tool exists to remove: a person, often a child, sent into the water to free a net. The design's own hazards are capsize from the pull on the pole, the operator falling back when the pin breaks, and a stronger substitute pin defeating the safety link; each has a control, and two need evidence from TRL 4 (the heel under a jerk, under the steady-lift rule of decision O6, and the pin calibration). The top project risk is that no co-design partner is found, without which nothing can be tested. Scales: likelihood and impact or severity 1 (low) to 5 (high); in the FMEA, detection 1 (sure to be found) to 5 (unlikely to be found) and RPN = severity x occurrence x detection. Method: STANDARDS section 23.

> **Safety:** Every control here assumes the standing rules: nobody enters the water, every person on the canoe wears a life jacket, the operator kneels, the crew slackens the line when the operator lifts and cuts the net rather than risk a capsize, and the tool is used only from canoes 7 m or longer with at least three crew.

Project risks

Table 1. Project risk register.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	No co-design partner	No NGO or landing committee approached yet	No staged-snag trials, no heel tests, no user validation	3	5	Project lead	Approach Challenging Heights and the Fisheries Commission's Volta zone office first (SMR-PLN-001)
P2	The tool is read as making children's work on canoes acceptable	Handed out without protection work, or marketed as a safety tool for divers	Harm to children; harm to the partner's work	2	5	Partner liaison	Distribution only through child-protection partners (D9); wording on every document and the tag (PR-013)
P3	Nets wrap more often than they hook	Snag statistics unknown	PR-001 not met; the hook head carries the load	3	4	Design lead	Ask partner crews first (DEC item to confirm 4); trial the hook head beside the ring
P4	Kit cost stays above USD 40	Prototype prices; galvanising minimum lots	PR-010 not met; fewer kits per canoe	4	3	Design lead	Local prices, batch galvanising, bamboo variant with shared crutch (13C)
P5	O-temper pin wire not available locally	Electrical wire is usually hard-drawn	Pins too strong; safety link defeated	3	5	Design lead	Specify, calibrate every reel (PR-011); anneal a reel and recalibrate if needed (to assess)
P6	Aluminium tube not available near the fishery	Supply	Aluminium prototype hard to repeat	3	2	Design lead	Bamboo local variant (SMR-DDR-003)

Preliminary hazard analysis

Table 2. Preliminary hazard analysis.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Capsize while working a snag	Pull on the pole heels the canoe; small canoe; standing operator	Drowning	5	2	Shear pin 302 N nominal; 3.4 deg static heel in the design canoe [H5]; 7 m and three-crew rule; kneel; life jackets; cut the net	Severity 5, likelihood 1 for steady pulls; jerk peak 6.5 deg [H11] ruled out by the steady-lift rule (O6, 2026-10-25) and measured at TRL 4; measured displacement x GM criterion (O9)
H2	Sudden release when the pin breaks	Pin shears at full pull	Operator falls back in the canoe; strikes crew with the pole top	3	3	Kneel; crew clear of the pole top; pole top painted orange	2, 3
H3	Stronger substitute pin	Nail, bolt or steel wire fitted when pins run out	Pull no longer capped; H1 returns	5	3	Eleven spare pins in the tube; rating marked on the pole head; training; PR-011	5, 2
H4	Someone enters the water anyway	Tool fails; habit	Drowning; the problem the tool exists to stop	5	2	Cut the net rather than dive; distribution with protection work; tag (PR-013)	5, 2
H5	Pinch at the ring gate and hinge	Closing the ring on the line	Finger injury	2	3	Gate pin on a lanyard; ring closed before it is near the line under tension	2, 2
H6	Hook head catches a hand or the operator's clothing	J hook near the canoe	Laceration; pulled toward the water	3	2	Hook head used kneeling, line slack, with the same pin	3, 1
H7	Welding fumes from galvanised steel	Welding after galvanising	Metal fume fever	3	2	Weld before galvanising, in the open (SMR-BLD-001 S1)	3, 1
H8	Crutch clamp slips off the gunwale	Loose clamp; gunwale outside 30 to 60 mm	Line jumps; crutch lost overboard	2	2	Clamp checked by hand before use (S3)	2, 1

Design FMEA (first pass, option A)

Table 3. Design FMEA.

Item	Function	Failure mode	Effect	Cause	Severity	Occurrence	Detection	RPN	Action
Shear pin (5)	Break at 256 to 347 N	Breaks high	Canoe heels further; H1	Wrong temper or alloy; thick wire	5	3	3	45	CalRig per reel (PR-011); O-temper specified; rating mark
Shear pin (5)	Break at 256 to 347 N	Breaks low	Pin breaks on hooked nets; slow use	Thin wire; nicked pin	2	3	3	18	Calibration; deburr; spares
Pole and pole head (4, 6, 7)	Carry push and pull to the ring	Pole buckles before the pin breaks	Pole bent; pull not capped cleanly	Factor 2.21 [S2]; damaged section	3	2	2	12	Replace bent sections; sleeves stiffen joints
Hook head (21)	Drag a wrapped bight back	J opens	Hook lost; bight not freed	10 mm bar at factor 1.86	3	1	2	6	Fixed by F1: 12 mm bar, factor 3.1 [S12]
Rider ring (1)	Close round the line and slide	Ring out of round; gate pin will not go in	Ring cannot be closed	Bending error over 2 mm	2	3	1	6	Bend over a former; check against a template (S17)
Gate pin (3)	Keep the ring closed	R-clip lost; pin falls out	Ring opens; ring lost if the tether fails	Lanyard cut	3	2	2	12	Lanyard; tether holds the ring
Tether and float (11, 12)	Recover the ring; float it if dropped	Bowline slips; foam waterlogged	Ring lost	Poor knot; damaged foam	2	2	2	8	Bowline checked; closed-cell foam; reserve 2.1 [B3]
Lock pins and rivets (8, 9)	Hold the pole joints	Corrosion of aluminium round stainless	Joint loose; pole separates	Galvanic couple in wet storage (S12)	3	2	3	18	Six month wet storage check (PR-008)
Galvanised steel parts	Resist wet storage	Coating damaged at	Rust	Drilling or grinding	2	2	2	8	All cutting before galvanising;

Item	Function	Failure mode	Effect	Cause	Severity	Occurrence	Detection	RPN	Action
		cuts after galvanising		after galvanising					zinc paint on damage
Bamboo pole (V1, V2)	Same as the aluminium pole	Culm splits or rots; buckles at 1.0 factor at low E	Pole fails; pull not capped cleanly	Untreated or weak culm	3	3	2	18	Bend test per culm [V10]; borax treatment; field check
Gunwale crutch (15 to 17)	Guide the line over the gunwale	Roller seizes	Line chafes	Grit in the bore	1	3	1	3	Rinse; HDPE on a bolt
Operator on the canoe	Lift steadily	Jerks the pole	Peak heel 6.5 deg [H11]	Snag lets go, frustration	4	3	3	36	Steady-lift rule in the operating rules and on the tag (O6, decided 2026-10-25); measured at TRL 4; 1.7 mm pin only if the test shows a real exceedance